Modelagem do Sistema de Injeção

Lista de Abreviações e Siglas

Abreviação	**Significado**
AC	Água de Resfriamento - Área Classificada
AD	Água Destilada
ADO	Água para Diluição de Óleo
AF	Água Fresca
AFDO	Água Fresca de Diluição de Óleo
AI	Anti-incrustante
AL	Alcalino
AM	Água do Mar
ANC	Água de Resfriamento - Área Não Classificada
AQ	Água Quente
B	Bomba
BB	Bomba Booster
BC	Bomba de Captação
BD	Biodispersante
BI	Biocida
BP	Bypass
CAP	Captação
DR	Dreno
DAM	Distribuição de Água do Mar
DAR	Distribuição de Água de Resfriamento
DES	Desaeradora
DOS	Dosadora
EC	Eletrocloradora
FF	Filtro Fino
FG	Filtro Grosso
FT	Filtro
GI	Gás Inerte
HC	Hipoclorito
INJ	Injeção
INT	Interconexões
LIM	Limpeza
LAV	Estações de Lavagem
LINHAAF	Linha de Água Fresca
LINHAAD	Linha de Água Destilada
LQ	Limpeza Química
LQDO	Limpeza Química - Diluição de Óleo
MIS	Mistura
MK	Tanque Makeup
NaHSO3	Bissulfito de Sódio
OB	Overboard
OD	Open Drain
P	Permeado
R	Retorno
SEQ	Sequestrante

Abreviação	**Significado**
SK	Skid
SL	Safe Location
SO	Sequestrante de Oxigênio
SR	Sistema de Resfriamento
TA, TB, TC, TD, TE	Trem A, B, C, D, E
TC	Trocador de Calor
TOT	
TQ	Tanque
URS	Unidade de Remoção de Sulfatos
UTI	Utilidades
VAC	
VE	Vessel
VENT	Vent

Variáveis calculadas

$F_{4,out,AD,UTI}$

$F_{4,out,AF,UTI}$

$F_{5,out,DAM,UTI}$

$F_{6,out,AC,UTI}$

1 Captação de Água do Mar

1.1 Eletrocloradora

$$F_{7,in,AM} = F_{7,in,BB_ABC} \tag{1}$$

$$F_{7,in,BB_ABC} = F_{7,out,BB_ABC} \tag{2}$$

$$\boldsymbol{F_{7,out,BB_ABC}} = F_{7,out,BB_ABC,MIS_1} \tag{3}$$

$$F_{7,out,BB_ABC,MIS_1} = F_{7,in,SR} + F_{7,in,EC} \tag{4}$$

$$F_{7,in,SR} = \boldsymbol{F_{7,out,SR,DR}} \tag{5}$$

$$F_{7,in,EC} = F_{7,in,EC_1} + F_{7,in,EC_2} \tag{6}$$

$$F_{7,in,EC_1} = F_{7,out,EC_1} \tag{7}$$

$$F_{7,in,EC_2} = F_{7,out,EC_2} \tag{8}$$

$$F_{7,in,TQ_HC,EC} = F_{7,out,EC_1} + F_{7,out,EC_2} \tag{9}$$

1.2 Tanque de Hipoclorito

$$F_{7,in,B_ABC} = F_{7,in,TQ_HC,EC} + F_{7,in,TQ_HC,B_ABC} \tag{10}$$

$$F_{7,in,B_ABC} = F_{7,in,TQ_HC,B_ABC} + \boldsymbol{F_{7,in,INJ,B_ABC}} \tag{11}$$

1.3 Captação de Água do Mar e Injeção de Hipoclorito

$$F_{7,in,CAP,INJ_HC1} = F_{7,in,BC_A,HC1} \tag{12}$$

$$F_{7,out,BC_A} = F_{7,in,BC_A,HC1} + F_{7,in,BC_A,AM} \tag{13}$$

$$F_{7,in,CAP,INJ_HC2} = F_{7,in,BC_B,HC2} \tag{14}$$

$$F_{7,out,BC_B} = F_{7,in,BC_B,HC2} + F_{7,in,BC_B,AM} \tag{15}$$

$$F_{7,in,CAP,INJ_HC3} = F_{7,in,BC_C,HC3} \tag{16}$$

$$F_{7,out,BC_C} = F_{7,in,BC_C,HC3} + F_{7,in,BC_C,AM} \tag{17}$$

$$F_{7,in,CAP,INJ_HC4} = F_{7,in,BC_D,HC4} \tag{18}$$

$$F_{7,out,BC_D} = F_{7,in,BC_D,HC4} + F_{7,in,BC_D,AM} \tag{19}$$

$$F_{7,in,CAP,INJ_HC5} = F_{7,in,BC_E,HC5} \tag{20}$$

$$F_{7,out,BC_E} = F_{7,in,BC_E,HC5} + F_{7,in,BC_E,AM} \tag{21}$$

$$F_{7,out,BC_A} + F_{7,out,BC_B} + F_{7,out,BC_C} + F_{7,out,BC_D} = F_{6,in,F_ABCD,BC_ABCD} \tag{22}$$

2 Distribuição de Água

$$F_{5,in,DAM} = F_{6,out,TC,DAM_LAV} + F_{6,out,F_ABCDE,DAM} \tag{23}$$

$$F_{5,in,DAM} = F_{5,in,FG,DAM} + F_{8,in,DES_SK,DAM} + F_{5,out,DAM,UTI} \tag{24}$$

2.1 Filtros - Distribuição

$$F_{6,in,F_ABCD,BC_ABCD} + F_{6,in,F_E,BC_E} = F_{6,in,TC,F_ABCD} + F_{6,in,ANC,F_ABCDE} + F_{6,out,F_ABCD,DAM} + F_{6,out,F_ABCD,VAC_1} + F_{6,out,F_ABCD,VAC_2} \tag{25}$$

$$F_{6,in,F_A} = F_{6,out,F_A} \tag{26}$$

$$F_{6,in,F_B} = F_{6,out,F_B} \tag{27}$$

$$F_{6,in,F_C} = F_{6,out,F_C} \tag{28}$$

$$F_{6,in,F_D} = F_{6,out,F_D} \tag{29}$$

$$F_{6,in,F_E} = F_{6,out,F_E} \tag{30}$$

2.2 Trocadores de Calor

$$F_{6,in,TC,F_ABCD} + F_{6,in,TC,INT_INJ_BB} + F_{6,in,TC,AC} = F_{6,out,TC,DAR} + F_{6,in,AC,TC} + F_{6,out,TC,DAM_LAV} + F_{8,in,INT_INJ_B,TC} \tag{31}$$

$$F_{6,in,TC_A1} + F_{6,in,TC_A2} = F_{6,out,TC_A} \tag{32}$$

$$F_{6,in,TC_B1} + F_{6,in,TC_B2} = F_{6,out,TC_B} \tag{33}$$

$$F_{6,in,TC_C1} + F_{6,in,TC_C2} = F_{6,out,TC_C} \tag{34}$$

$$F_{6,in,TC_D1} + F_{6,in,TC_D2} = F_{6,out,TC_D} \tag{35}$$

$$F_{6,in,TC_E1} + F_{6,in,TC_E2} = F_{6,out,TC_E} \tag{36}$$

2.3 Água de Refrigeração - AC

$$F_{6,in,AC,DAR_1} + F_{6,in,AC,DAR_2} + F_{6,in,AC,TC} + F_{6,in,AC,AF_MK} + F_{6,in,AC,GI} = F_{6,in,TC,AC} + F_{6,out,AC,UTI} + F_{6,out,AC,SL_1} + F_{6,out,AC,SL_2} \tag{37}$$

$$F_{6,in,B_A1} = F_{6,out,B_A1} \tag{38}$$

$$F_{6,in,B_B1} = F_{6,out,B_B1} \tag{39}$$

$$F_{6,in,B_C1} = F_{6,out,B_C1} \tag{40}$$

$$F_{6,in,B_A2} = F_{6,out,B_A2} \tag{41}$$

$$F_{6,in,B_B2} = F_{6,out,B_B2} \tag{42}$$

$$F_{6,in,F} = F_{6,out,F} \tag{43}$$

$$F_{6,in,VE1_1} + F_{6,in,VE1_2} + F_{6,in,VE1_3} = F_{6,in,SL_1} + F_{6,in,SL_2} + F_{6,out,VE1} \tag{44}$$

$$F_{6,in,VE2} = F_{6,out,VE2_1} + F_{6,out,VE2_2} \tag{45}$$

2.4 Água de Refrigeração - ANC

$$F_{6,in,ANC,F_ABCDE} + F_{6,in,ANC,B_AQ} + F_{6,in,ANC,DAR} = F_{6,in,TC,ANC} + F_{6,out,ANC,DAR} + F_{6,out,ANC,B_AQ} \tag{46}$$

$$F_{6,in,B_A} = F_{6,out,B_A} \tag{47}$$

$$F_{6,in,B_B} = F_{6,out,B_B} \tag{48}$$

$$F_{6,in,B_C} = F_{6,out,B_C} \tag{49}$$

$$F_{6,in,TC_A1} + F_{6,in,TC_A2} = F_{6,out,TC_A1} + F_{6,out,TC_A2} \tag{50}$$

$$F_{6,in,TC_B1} + F_{6,in,TC_B2} = F_{6,out,TC_B1} + F_{6,out,TC_B2} \tag{51}$$

$$F_{6,in,VE} = F_{6,out,VE} \tag{52}$$

$$F_{6,in,TQ} = F_{6,out,TQ_1} + F_{6,out,TQ_2} \tag{53}$$

3 Tratamento da Água

3.1 Filtros Grossos

$$F_{1,in,FG,DAM} + F_{1,in,FG,BI_SK1} = F_{1,in,FF,FG} \tag{54}$$

$$F_{1,in,FG_A} = F_{1,out,FG_A} \tag{55}$$

$$F_{1,in,FG_B} = F_{1,out,FG_B} \tag{56}$$

$$F_{1,in,FG_C} = F_{1,out,FG_C} \tag{57}$$

$$F_{1,in,FF,FG} = F_{1,out,FG_A} + F_{1,out,FG_B} + F_{1,out,FG_C} \tag{58}$$

3.2 Filtros Finos

$$F_{1,in,FF,FG} + F_{1,in,FF,BI_SK1} + F_{1,in,FF,NaHSO3_SK3} + \boldsymbol{F_{1,in,FF,AI_SK2}} = F_{1,out,FF,AF} + F_{1,in,BB,FF} \tag{59}$$

$$F_{1,in,FF_A} = F_{1,out,FF_A} \tag{60}$$

$$F_{1,in,FF_B} = F_{1,out,FF_B} \tag{61}$$

$$F_{1,in,FF_C} = F_{1,out,FF_C} \tag{62}$$

$$F_{1,in,BB,FF} = F_{1,out,FF_A} + F_{1,out,FF_B} + F_{1,out,FF_C} \tag{63}$$

3.3 Bomba Booster

$$F_{1,in,BB,FF} = F_A + F_B + F_C + F_D + F_E + F_{1,in,URS_BP,BB} \tag{64}$$

$$\boldsymbol{F_{1,out,BB,URS}} = F_{1,in,URS_BP,BB} \tag{65}$$

$$F_{1,in,BB,FF} = F_{1,in,BB_A} + F_{1,in,BB_B} \tag{66}$$

$$F_{1,in,BB_A} = F_{1,out,BB_A} \tag{67}$$

$$F_{1,in,BB_B} = F_{1,out,BB_B} \tag{68}$$

3.4 Sistema de Membranas da URS

3.4.1 Reconciliação

$$F = \boldsymbol{R} + P \tag{69}$$

$$P = P_A + P_B + P_C + P_D + P_E \tag{70}$$

$$P_A = \boldsymbol{P_{a,A}} + \boldsymbol{P_{b,A}} + \boldsymbol{P_{c,A}} \tag{71}$$

$$P_B = \boldsymbol{P_{a,B}} + \boldsymbol{P_{b,B}} + \boldsymbol{P_{c,B}} \tag{72}$$

$$P_C = \boldsymbol{P_{a,C}} + \boldsymbol{P_{b,C}} + \boldsymbol{P_{c,C}} \tag{73}$$

$$P_D = \boldsymbol{P_{a,D}} + \boldsymbol{P_{b,D}} + \boldsymbol{P_{c,D}} \tag{74}$$

$$P_E = \boldsymbol{P_{a,E}} + \boldsymbol{P_{b,E}} + \boldsymbol{P_{c,E}} \tag{75}$$

$$\boldsymbol{R} = \boldsymbol{R_{c,A}} + \boldsymbol{R_{c,B}} + \boldsymbol{R_{c,C}} + \boldsymbol{R_{c,D}} + \boldsymbol{R_{c,E}} \tag{76}$$

$$F = F_A + F_B + F_C + F_D + F_E \tag{77}$$

$$F_A = \boldsymbol{R_{a,A}} + \boldsymbol{P_{a,A}} + R_{b,A} + \boldsymbol{P_{b,A}} \tag{78}$$

$$F_B = \boldsymbol{R_{a,B}} + \boldsymbol{P_{a,B}} + R_{b,B} + \boldsymbol{P_{b,B}} \tag{79}$$

$$F_C = \boldsymbol{R_{a,C}} + \boldsymbol{P_{a,C}} + R_{b,C} + \boldsymbol{P_{b,C}} \tag{80}$$

$$F_D = \boldsymbol{R_{a,D}} + \boldsymbol{P_{a,D}} + R_{b,D} + \boldsymbol{P_{b,D}} \tag{81}$$

$$F_E = \boldsymbol{R_{a,E}} + \boldsymbol{P_{a,E}} + R_{b,E} + \boldsymbol{P_{b,E}} \tag{82}$$

$$R_{b,A} = F_{c,A} - \boldsymbol{R_{a,A}} = \boldsymbol{R_{c,A}} + \boldsymbol{P_{c,A}} - \boldsymbol{R_{a,A}} \tag{83}$$

$$R_{b,B} = F_{c,B} - \boldsymbol{R_{a,B}} = \boldsymbol{R_{c,B}} + \boldsymbol{P_{c,B}} - \boldsymbol{R_{a,B}} \tag{84}$$

$$R_{b,C} = F_{c,C} - \boldsymbol{R_{a,C}} = \boldsymbol{R_{c,C}} + \boldsymbol{P_{c,C}} - \boldsymbol{R_{a,C}} \tag{85}$$

$$R_{b,D} = F_{c,D} - \boldsymbol{R_{a,D}} = \boldsymbol{R_{c,D}} + \boldsymbol{P_{c,D}} - \boldsymbol{R_{a,D}} \tag{86}$$

$$R_{b,E} = F_{c,E} - \boldsymbol{R_{a,E}} = \boldsymbol{R_{c,E}} + \boldsymbol{P_{c,E}} - \boldsymbol{R_{a,E}} \tag{87}$$

$$F_{8,in,DES,URS} = P + F_{1,in,URS_BP,BB} \tag{88}$$

3.4.2 Monitoramento por Estágios

Algumas variáveis representam o mesmo ponto físico, mas recebem nomes diferentes em cada contexto (estágio / banco). Adota-se um nome canônico único para cada ponto, de forma a evitar redundância no sistema de equações. A Tabela **??** apresenta o mapeamento adotado.

Nome canônico	Nomes equivalentes
PT_{RF}	$PT_{1R} = PT_{2F} = PT_{cF}$
PT_R	$PT_{cR} = PT_{2R}$
PT_P	$PT_{2P} = PT_{cP}$
PDI_{c2}	$PDI_c = PDI_2$

$$S_{1,A} = \frac{\dfrac{\boldsymbol{P_{a,A}} + \boldsymbol{P_{b,A}}}{\boldsymbol{A}}}{\boldsymbol{PT_{RF,A}} - PT_{1P,A} - \boldsymbol{\Delta\pi}} \tag{89}$$

$$S_{1,B} = \frac{\dfrac{\boldsymbol{P_{a,B}} + \boldsymbol{P_{b,B}}}{\boldsymbol{A}}}{\boldsymbol{PT_{RF,B}} - PT_{1P,B} - \boldsymbol{\Delta\pi}} \tag{90}$$

$$S_{1,C} = \frac{\dfrac{\boldsymbol{P_{a,C}} + \boldsymbol{P_{b,C}}}{\boldsymbol{A}}}{\boldsymbol{PT_{RF,C}} - PT_{1P,C} - \boldsymbol{\Delta\pi}} \tag{91}$$

$$S_{1,D} = \frac{\frac{\boldsymbol{P_{a,D}} + \boldsymbol{P_{b,D}}}{\boldsymbol{A}}}{\boldsymbol{PT_{RF,D}} - PT_{1P,D} - \boldsymbol{\Delta\pi}} \tag{92}$$

$$S_{1,E} = \frac{\frac{\boldsymbol{P_{a,E}} + \boldsymbol{P_{b,E}}}{\boldsymbol{A}}}{\boldsymbol{PT_{RF,E}} - PT_{1P,E} - \boldsymbol{\Delta\pi}} \tag{93}$$

$$S_{2,A} = \frac{\frac{\boldsymbol{P_{c,A}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,A}} - PT_{P,A} - \boldsymbol{\Delta\pi}} \tag{94}$$

$$S_{2,B} = \frac{\frac{\boldsymbol{P_{c,B}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,B}} - PT_{P,B} - \boldsymbol{\Delta\pi}} \tag{95}$$

$$S_{2,C} = \frac{\frac{\boldsymbol{P_{c,C}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,C}} - PT_{P,C} - \boldsymbol{\Delta\pi}} \tag{96}$$

$$S_{2,D} = \frac{\frac{\boldsymbol{P_{c,D}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,D}} - PT_{P,D} - \boldsymbol{\Delta\pi}} \tag{97}$$

$$S_{2,E} = \frac{\frac{\boldsymbol{P_{c,E}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,E}} - PT_{P,E} - \boldsymbol{\Delta\pi}} \tag{98}$$

$$IDF_{1,A} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{RF,A}} - PT_{1P,A} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,A}} + \boldsymbol{P_{b,A}}}{\boldsymbol{A}} \tag{99}$$

$$IDF_{1,B} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{RF,B}} - PT_{1P,B} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,B}} + \boldsymbol{P_{b,B}}}{\boldsymbol{A}} \tag{100}$$

$$IDF_{1,C} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{RF,C}} - PT_{1P,C} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,C}} + \boldsymbol{P_{b,C}}}{\boldsymbol{A}} \tag{101}$$

$$IDF_{1,D} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{RF,D}} - PT_{1P,D} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,D}} + \boldsymbol{P_{b,D}}}{\boldsymbol{A}} \tag{102}$$

$$IDF_{1,E} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{RF,E}} - PT_{1P,E} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,E}} + \boldsymbol{P_{b,E}}}{\boldsymbol{A}} \tag{103}$$

$$IDF_{2,A} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{R,A}} - PT_{P,A} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,A}}}{\boldsymbol{A}} \tag{104}$$

$$IDF_{2,B} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{R,B}} - PT_{P,B} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,B}}}{\boldsymbol{A}} \tag{105}$$

$$IDF_{2,C} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{R,C}} - PT_{P,C} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,C}}}{\boldsymbol{A}} \tag{106}$$

$$IDF_{2,D} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{R,D}} - PT_{P,D} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,D}}}{\boldsymbol{A}} \tag{107}$$

$$IDF_{2,E} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{R,E}} - PT_{P,E} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,E}}}{\boldsymbol{A}} \tag{108}$$

$$\boldsymbol{PDI_{1,A}} = \boldsymbol{PT_{1F,A}} - \boldsymbol{PT_{RF,A}} \tag{109}$$

$$\boldsymbol{PDI_{1,B}} = \boldsymbol{PT_{1F,B}} - \boldsymbol{PT_{RF,B}} \tag{110}$$

$$\boldsymbol{PDI_{1,C}} = \boldsymbol{PT_{1F,C}} - \boldsymbol{PT_{RF,C}} \tag{111}$$

$$\boldsymbol{PDI_{1,D}} = \boldsymbol{PT_{1F,D}} - \boldsymbol{PT_{RF,D}} \tag{112}$$

$$\boldsymbol{PDI_{1,E}} = \boldsymbol{PT_{1F,E}} - \boldsymbol{PT_{RF,E}} \tag{113}$$

$$\boldsymbol{PDI_{c2,A}} = \boldsymbol{PT_{RF,A}} - \boldsymbol{PT_{R,A}} \tag{114}$$

$$\boldsymbol{PDI_{c2,B}} = \boldsymbol{PT_{RF,B}} - \boldsymbol{PT_{R,B}} \tag{115}$$

$$\boldsymbol{PDI_{c2,C}} = \boldsymbol{PT_{RF,C}} - \boldsymbol{PT_{R,C}} \tag{116}$$

$$\boldsymbol{PDI_{c2,D}} = \boldsymbol{PT_{RF,D}} - \boldsymbol{PT_{R,D}} \tag{117}$$

$$\boldsymbol{PDI_{c2,E}} = \boldsymbol{PT_{RF,E}} - \boldsymbol{PT_{R,E}} \tag{118}$$

3.4.3 Monitoramento por Bancos

$$S_{a,A} = \frac{\frac{\boldsymbol{P_{a,A}}}{\boldsymbol{A}}}{PT_{aR,A} - PT_{aP,A} - \boldsymbol{\Delta\pi}} \tag{119}$$

$$S_{a,B} = \frac{\frac{\boldsymbol{P_{a,B}}}{\boldsymbol{A}}}{PT_{aR,B} - PT_{aP,B} - \boldsymbol{\Delta\pi}} \tag{120}$$

$$S_{a,C} = \frac{\frac{\boldsymbol{P_{a,C}}}{\boldsymbol{A}}}{PT_{aR,C} - PT_{aP,C} - \boldsymbol{\Delta\pi}} \tag{121}$$

$$S_{a,D} = \frac{\frac{\boldsymbol{P_{a,D}}}{\boldsymbol{A}}}{PT_{aR,D} - PT_{aP,D} - \boldsymbol{\Delta\pi}} \tag{122}$$

$$S_{a,E} = \frac{\frac{\boldsymbol{P_{a,E}}}{\boldsymbol{A}}}{PT_{aR,E} - PT_{aP,E} - \boldsymbol{\Delta\pi}} \tag{123}$$

$$S_{b,A} = \frac{\frac{\boldsymbol{P_{b,A}}}{\boldsymbol{A}}}{PT_{bR,A} - PT_{bP,A} - \boldsymbol{\Delta\pi}} \tag{124}$$

$$S_{b,B} = \frac{\frac{\boldsymbol{P_{b,B}}}{\boldsymbol{A}}}{PT_{bR,B} - PT_{bP,B} - \boldsymbol{\Delta\pi}} \tag{125}$$

$$S_{b,C} = \frac{\frac{\boldsymbol{P_{b,C}}}{\boldsymbol{A}}}{PT_{bR,C} - PT_{bP,C} - \boldsymbol{\Delta\pi}} \tag{126}$$

$$S_{b,D} = \frac{\frac{\boldsymbol{P_{b,D}}}{\boldsymbol{A}}}{PT_{bR,D} - PT_{bP,D} - \boldsymbol{\Delta\pi}} \tag{127}$$

$$S_{b,E} = \frac{\frac{\boldsymbol{P_{b,E}}}{\boldsymbol{A}}}{PT_{bR,E} - PT_{bP,E} - \boldsymbol{\Delta\pi}} \tag{128}$$

$$S_{c,A} = \frac{\frac{\boldsymbol{P_{c,A}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,A}} - PT_{P,A} - \boldsymbol{\Delta\pi}} \tag{129}$$

$$S_{c,B} = \frac{\frac{\boldsymbol{P_{c,B}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,B}} - PT_{P,B} - \boldsymbol{\Delta\pi}} \tag{130}$$

$$S_{c,C} = \frac{\frac{\boldsymbol{P_{c,C}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,C}} - PT_{P,C} - \boldsymbol{\Delta\pi}} \tag{131}$$

$$S_{c,D} = \frac{\frac{\boldsymbol{P_{c,D}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,D}} - PT_{P,D} - \boldsymbol{\Delta\pi}} \tag{132}$$

$$S_{c,E} = \frac{\frac{\boldsymbol{P_{c,E}}}{\boldsymbol{A}}}{\boldsymbol{PT_{R,E}} - PT_{P,E} - \boldsymbol{\Delta\pi}} \tag{133}$$

$$IDF_{a,A} = \boldsymbol{L_p} \cdot (PT_{aR,A} - PT_{aP,A} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,A}}}{\boldsymbol{A}} \tag{134}$$

$$IDF_{a,B} = \boldsymbol{L_p} \cdot (PT_{aR,B} - PT_{aP,B} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,B}}}{\boldsymbol{A}} \tag{135}$$

$$IDF_{a,C} = \boldsymbol{L_p} \cdot (PT_{aR,C} - PT_{aP,C} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,C}}}{\boldsymbol{A}} \tag{136}$$

$$IDF_{a,D} = \boldsymbol{L_p} \cdot (PT_{aR,D} - PT_{aP,D} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,D}}}{\boldsymbol{A}} \tag{137}$$

$$IDF_{a,E} = \boldsymbol{L_p} \cdot (PT_{aR,E} - PT_{aP,E} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{a,E}}}{\boldsymbol{A}} \tag{138}$$

$$IDF_{b,A} = \boldsymbol{L_p} \cdot (PT_{bR,A} - PT_{bP,A} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{b,A}}}{\boldsymbol{A}} \tag{139}$$

$$IDF_{b,B} = \boldsymbol{L_p} \cdot (PT_{bR,B} - PT_{bP,B} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{b,B}}}{\boldsymbol{A}} \tag{140}$$

$$IDF_{b,C} = \boldsymbol{L_p} \cdot (PT_{bR,C} - PT_{bP,C} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{b,C}}}{\boldsymbol{A}} \tag{141}$$

$$IDF_{b,D} = \boldsymbol{L_p} \cdot (PT_{bR,D} - PT_{bP,D} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{b,D}}}{\boldsymbol{A}} \tag{142}$$

$$IDF_{b,E} = \boldsymbol{L_p} \cdot (PT_{bR,E} - PT_{bP,E} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{b,E}}}{\boldsymbol{A}} \tag{143}$$

$$IDF_{c,A} = \boldsymbol{L_p} \cdot (PT_{R,A} - PT_{P,A} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,A}}}{\boldsymbol{A}} \tag{144}$$

$$IDF_{c,B} = \boldsymbol{L_p} \cdot (PT_{R,B} - PT_{P,B} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,B}}}{\boldsymbol{A}} \tag{145}$$

$$IDF_{c,C} = \boldsymbol{L_p} \cdot (PT_{R,C} - PT_{P,C} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,C}}}{\boldsymbol{A}} \tag{146}$$

$$IDF_{c,D} = \boldsymbol{L_p} \cdot (PT_{R,D} - PT_{P,D} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,D}}}{\boldsymbol{A}} \tag{147}$$

$$IDF_{c,E} = \boldsymbol{L_p} \cdot (PT_{R,E} - PT_{P,E} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_{c,E}}}{\boldsymbol{A}} \tag{148}$$

$$PDI_{a,A} = PT_{aF,A} - PT_{aR,A} \tag{149}$$

$$PDI_{a,B} = PT_{aF,B} - PT_{aR,B} \tag{150}$$

$$PDI_{a,C} = PT_{aF,C} - PT_{aR,C} \tag{151}$$

$$PDI_{a,D} = PT_{aF,D} - PT_{aR,D} \tag{152}$$

$$PDI_{a,E} = PT_{aF,E} - PT_{aR,E} \tag{153}$$

$$PDI_{b,A} = PT_{bF,A} - PT_{bR,A} \tag{154}$$

$$PDI_{b,B} = PT_{bF,B} - PT_{bR,B} \tag{155}$$

$$PDI_{b,C} = PT_{bF,C} - PT_{bR,C} \tag{156}$$

$$PDI_{b,D} = PT_{bF,D} - PT_{bR,D} \tag{157}$$

$$PDI_{b,E} = PT_{bF,E} - PT_{bR,E} \tag{158}$$

$$\boldsymbol{PDI_{c2,A} = PT_{RF,A} - PT_{R,A}} \tag{159}$$

$$\boldsymbol{PDI_{c2,B} = PT_{RF,B} - PT_{R,B}} \tag{160}$$

$$\boldsymbol{PDI_{c2,C} = PT_{RF,C} - PT_{R,C}} \tag{161}$$

$$\boldsymbol{PDI_{c2,D} = PT_{RF,D} - PT_{R,D}} \tag{162}$$

$$\boldsymbol{PDI_{c2,E} = PT_{RF,E} - PT_{R,E}} \tag{163}$$

3.5 Injeção de Biocida

$$F_{1,in,TQ_1,TOT_BI} = F_{1,in,FG,BI_SK1} + F_{1,in,FF,BI_SK1} \tag{164}$$

$$F_{1,in,TQ_1} = F_{1,in,SK1} \tag{165}$$

$$F_{1,in,SK1} = F_{1,in,FG,BI_SK1} + F_{1,in,FF,BI_SK1} \tag{166}$$

$$\boldsymbol{F_{1,out,SK1}} = F_{1,in,FG,BI_SK1} + F_{1,in,FF,BI_SK1} \tag{167}$$

3.6 Injeção de Anti-incrustante

$$F_{3,in,TQ_2,TOT_AI} = F_{1,in,FF,AI_SK2} \tag{168}$$

$$F_{3,in,TQ_2} = F_{3,in,SK2} \tag{169}$$

$$F_{3,in,SK2} = \boldsymbol{F_{1,in,FF,AI_SK2}} \tag{170}$$

3.7 Injeção de Bissulfito de Sódio

$$F_{3,in,TQ_3,TOT_NaHSO3} = F_{1,in,FF,NaHSO3_SK3} + F_{4,in,LIM,NaHSO3_SK3} \tag{171}$$

$$F_{3,in,TQ_3} = F_{3,in,SK3} \tag{172}$$

$$F_{3,in,SK3} = F_{1,in,FF,NaHSO3_SK3} + F_{4,in,LIM,NaHSO3_SK3} \tag{173}$$

$$\boldsymbol{F_{3,out,SK3}} = F_{1,in,FF,NaHSO3_SK3} + F_{4,in,LIM,NaHSO3_SK3} \tag{174}$$

3.8 Sistema de Limpeza

$$F_{4,in,LIM,AD} + F_{4,in,LIM,NaHSO3_SK3} = F_{4,in,MIS_1} + F_{4,in,MIS_2} \tag{175}$$

$$F_{4,in,RLQ,TABCDE} + F_{4,in,RLQ,AFDO} = F_{4,in,MIS_3} + F_{4,in,MIS_4} \tag{176}$$

$$F_{4,in,TQ_A,AL} + F_{4,in,TQ_A,AC} + F_{4,in,MIS_1} + F_{4,in,MIS_3} = F_{4,out,TQ_A} \tag{177}$$

$$\begin{aligned} &F_{4,in,TQ_B,AL} + F_{4,in,TQ_B,AC} + F_{4,in,MIS_2} + F_{4,in,MIS_4} \\ &\quad + F_{4,in,TQ_B,B_ABC} = F_{4,out,TQ_B} \end{aligned} \tag{178}$$

$$F_{4,out,TQ_A} + F_{4,out,TQ_B} = F_{4,in,B_A} + F_{4,in,B_B} + F_{4,in,B_C} \tag{179}$$

$$F_{4,in,B_A} = F_{4,out,B_A} \tag{180}$$

$$F_{4,in,B_B} = F_{4,out,B_B} \tag{181}$$

$$F_{4,in,B_C} = F_{4,out,B_C} \tag{182}$$

$$F_{4,out,B_A} + F_{4,out,B_B} + F_{4,out,B_C} = F_{4,in,TQ_B,B_ABC} + F_{4,in,F_3} \tag{183}$$

$$F_{4,in,F_3} = \boldsymbol{F_{4,out,F_3}} \tag{184}$$

$$\boldsymbol{F_{4,out,F_3}} = F_{4,in,LQ_TABCDE,FT3} + F_{4,in,LQDO_TA,FT3} + F_{4,in,LQDO_TB,FT3} \tag{185}$$

$$F_{4,in,LINHAAF} = F_{4,out,AF,UTI} + F_{4,in,ANC,AF} + F_{4,in,UTI_AQ,AF} + F_{4,in,AF_MK,AF} \tag{186}$$

$$F_{4,in,AF_MK,AF} = F_{4,out,AF_MK,B_AQ} + F_{6,in,AC,AF_MK} \tag{187}$$

$$F_{4,in,LINHAAD} + F_{4,in,SO} = F_{4,out,AD,UTI} + F_{4,in,LIM,AD} \tag{188}$$

4 Injeção de Água no Poço

4.1 Desaeração a Vácuo

$$F_{8,in,DES,URS} + F_{8,in,DES,BI} = \boldsymbol{F_{8,in,DES,URS_BI}} + F_{8,in,INJ,URS_BI} \tag{189}$$

$$F_{8,out,DES_G} = F_{8,out,DES_G1} + F_{8,out,DES_G2} + F_{8,out,VENT} \tag{190}$$

$$\boldsymbol{F_{8,in,DES,URS_BI}} + F_{8,in,DES,INJ} + F_{8,in,DES,SE} = F_{8,out,DES_1} + F_{8,out,DES_2} + F_{8,out,DES_G} \tag{191}$$

$$F_{8,in,INJ,DES} = F_{8,out,DES_1} + F_{8,in,INJ,URS_BI} + F_{8,in,INJ,BI} + F_{8,in,INJ,SE} + F_{8,in,INJ,BD} \tag{192}$$

$$F_{8,out,SK,FF_OB} = F_{8,in,SK,G_1} + F_{8,in,SK,G_2} + F_{8,in,DES_SK,DAM} + F_{8,in,SK,AR} \tag{193}$$

4.2 Bombeamento de Água de Injeção

$$F_{8,in,INJ_BB,DES} = F_{8,in,BB_A} + F_{8,in,BB_B} + F_{8,in,BB_C} \tag{194}$$

$$F_{8,in,BB_A} = F_{8,out,BB_A} \tag{195}$$

$$F_{8,in,BB_B} = F_{8,out,BB_B} \tag{196}$$

$$F_{8,in,BB_C} = F_{8,out,BB_C} \tag{197}$$

$$F_{8,out,BB_A} + F_{8,out,BB_B} + F_{8,out,BB_C} = F_{6,in,TC,INT_INJ_BB} + F_{8,out,BB,URS} \tag{198}$$

$$F_{8,in,INT_INJ_B,TC} = F_{8,in,B_A,TC} + F_{8,in,B_B,TC} \tag{199}$$

$$\boldsymbol{F_{8,in,B_A,TC}} + F_{8,in,B_A,DAR} = F_{8,out,B_A} + F_{8,out,B_A,ANC} \tag{200}$$

$$\boldsymbol{F_{8,in,B_B,TC}} + F_{8,in,B_B,DAR} = F_{8,out,B_B} + F_{8,out,B_B,ANC} \tag{201}$$

$$\boldsymbol{F_{8,out,B_AB}} = F_{8,out,B_A} + F_{8,out,B_B} \tag{202}$$

$$\boldsymbol{F_{8,out,B_AB}} = F_{8,in,AI,IJN} + F_{8,in,ADO,INJ} \tag{203}$$